Sports Biomechanics

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rspb20

Lower extremity kinematic and kinetic factors associated with bat speed at ball contact during the baseball swing

Karl F. Orishimo, Ian J. Kremenic, Edward Modica Jr., Takumi Fukunaga, Malachy P. McHugh & Srino Bharam

To cite this article: Karl F. Orishimo, Ian J. Kremenic, Edward Modica Jr., Takumi Fukunaga, Malachy P. McHugh & Srino Bharam (18 Oct 2023): Lower extremity kinematic and kinetic factors associated with bat speed at ball contact during the baseball swing, Sports Biomechanics, DOI: 10.1080/14763141.2023.2269418

To link to this article: https://doi.org/10.1080/14763141.2023.2269418

Published online: 18 Oct 2023.

SPORTS BIOMECHANICS
https://doi.org/10.1080/14763141.2023.2269418

Lower extremity kinematic and kinetic factors associated with bat speed at ball contact during the baseball swing

Karl F. Orishimo[a], Ian J. Kremenic[a], Edward Modica Jr.[b], Takumi Fukunaga[a], Malachy P. McHugh[a] and Srino Bharam[c]

[a]Nicholas Institute of Sports Medicine and Athletic Trauma, Manhattan, Eye, Ear and Throat Hospital, Northwell Health, New York, NY, USA; [b]College of Osteopathic Medicine, New York Institute of Technology, Old Westbury, NY, USA; [c]Department of Orthopaedics, Lenox Hill Hospital, Northwell Health, New York, NY, USA

ABSTRACT

The purpose of this study was to determine which biomechanical variables measured during the baseball swing are associated with linear bat speed at ball contact (bat speed). Twenty collegiate baseball players hit a baseball from a tee into a net. Kinematics were recorded with a motion capture system sampling at 500 Hz and kinetics were measured by force plates under each foot sampling at 1000 Hz. Associations between bat speed, individual joint and segment kinematics, joint moments and ground reaction forces (GRF) were assessed using Pearson correlations and stepwise linear regression. Average bat speed was 30 ± 2 m/s. Lead foot peak vertical (159 ± 29% BW, $r = 0.622$, $P = 0.001$), posterior (–57 ± 12% BW, $r = –0.574$, $P = 0.008$) and resultant (170 ± 30% BW, $r = 0.662$, $P = 0.001$) GRF were all correlated with bat speed. No combination of factors strengthened the relationship to bat speed beyond these individual variables. These results illustrate the role of the lead leg in generating and transferring ground reaction forces through the kinetic chain in order to accelerate the bat. Training to improve bat speed should include both general lower extremity strengthening exercises and sport-specific hitting drills to improve lower extremity force production following lead foot contact.

ARTICLE HISTORY
Received 30 August 2022
Accepted 18 September 2023

KEYWORDS
Hitting; lead leg; ground reaction force; hip-shoulder separation; correlation

Introduction

According to baseball great Ted Williams, hitting a baseball is 'the single most difficult thing to do in sports' (Williams & Underwood, 1970). When thrown from regulation distance from the pitchers' mound to home plate (18.4 m), a pitch travelling 90 mph (40.2 m/s) reaches the hitter in about 420 ms. During this time, the hitter must recognise the speed and trajectory of the pitch and then swing the bat such that it impacts the ball at the proper location to hit it in the field of play. Therefore, the ability to move the bat at high speed is an important characteristic of a skilled hitter. Hitting the ball with greater bat speed will transfer more momentum to the ball resulting in a higher ball exit velocity,

CONTACT Karl F. Orishimo karl@nismat.org
Study performed at Nicholas Institute of Sports Medicine and Athletic Trauma, Manhattan Eye, Ear and Throat Hospital, New York, NY, USA.

which is a highly valued metric when assessing player potential and development (Inkster et al., 2011). Additionally, a faster bat speed affords the hitter with more time to recognise the pitch type and its location and to decide whether or not to swing.

The baseball swing has been described as ‘a fluid sequential motion involving two movements working in tandem—straight forward or linear then angular or rotation’ (DeRenne, 2011). Following lead foot contact with the ground, the hitter’s weight is shifted forward from the trail leg to the lead leg (Welch et al., 1995). As the lead leg is loaded, energy is transferred through the kinetic chain. Ideally, the pelvis rotates towards the pitcher first followed by the trunk, upper extremity and finally the bat (Welch et al., 1995). The angular ‘separation’, or countermovement between adjacent segments, stretches the active muscles between segments and activates the stretch-shortening cycle (Putnam, 1993). The release of elastic energy during the ensuing concentric contraction increases the force and power output of the muscle, which results in the maximum rotational velocity of each segment being greater than that of its predecessor (Putnam, 1993; Welch et al., 1995).

The downward phase of the golf swing incorporates similar movement patterns (i.e., weight-shift and transverse rotation) to the bat acceleration phase of hitting. As generating high clubhead speeds is essential to driving the ball long distances in golf, kinematic and kinetic features of the golf swing that may influence clubhead speed have been thoroughly investigated. Previous studies have demonstrated significant associations between linear clubhead speed and biomechanical features of the golf swing such as, peak hip-shoulder separation, peak transverse plane trunk rotational velocity (Joyce, 2017), peak lead leg ground reaction forces (Han et al., 2019) and the work generated at the individual joints of the lead leg (McNally et al., 2014).

While the relationships between golf swing biomechanics and clubhead speed have been well documented, associations between hitting biomechanics and bat speed have not been extensively examined. In an electromyographic study, Shaffer et al. (1993) found that the high activation levels in the hamstring, gluteus maximus and quadriceps muscles of the trail leg contributed to the generation of power to rotate the segments, as well as provided a stable base of support. These authors, however, did not provide any kinetic or kinematic information regarding swing mechanics and did not examine muscle activity in the lead leg. Additionally, while greater linear bat speeds have been found in hitters with greater lead knee, pelvis and trunk angular velocities (Escamilla et al., 2009; Inkster et al., 2011), correlation analyses between bat speed and swing mechanics have not been explicitly performed. Furthermore, lead knee extension from lead foot contact to ball contact has been speculated to aid in generating the necessary force to stabilise the lower extremity and to drive the pelvis into rotation (Dowling & Fleisig, 2016; Escamilla et al., 2009; Welch et al., 1995), beginning the kinematic sequence. However, neither ground reaction forces nor joint moments from the lead or trail legs have been previously examined and the relationships between bat speed and these kinetic variables remain unclear.

While the specific relationships between ground reaction force production, kinematic sequencing, segment separation and bat speed have not been studied scientifically, numerous coaches and hitting instructors have emphasised them as major components of a powerful and efficient swing (DeRenne, 2011; Lau & Glossbrenner, 1984; Williams & Underwood, 1970). Gaining insight into the biomechanical factors that may be associated with bat speed may aid in the development of sport-specific training programs targeted at improving this important performance metric. Therefore, the purpose of this study was to investigate the relationships between kinematic (hip-shoulder separation and peak transverse plane pelvis and trunk velocities) and kinetic (lead and trail leg ground reaction forces and joint moments) factors and linear bat speed at ball contact (hereafter referred to as bat speed) during the baseball swing. Based on previous research regarding both baseball and golf swing biomechanics, it was hypothesised that pelvis and trunk kinematics, as well as the ground reaction forces and joint moments of the lead leg, would be correlated with bat speed.

Materials and methods

Twenty collegiate baseball players (age: 21 ± 2 yr, ht.: 180 ± 5 cm, wt.: 87.0 ± 9.0 kg) stood with each foot on separate force plates and hit a standard baseball from a tee into a net using a 32-inch (81.3 cm), 29-ounce (0.82 kg), aluminium baseball bat. All hitters were healthy and injury-free at the time of testing and were being examined as part of a pre-season fitness evaluation. This study was approved by the Northwell Health Institutional Review Board (Study# 18–0130) and each participant gave written informed consent to participate.

Thirty retro-reflective markers were placed on the lower extremities, arms and trunk. The marker locations were as follows: C7 vertebra, sacrum, and bilaterally on the acromia, medial epicondyle of the humerus, lateral epicondyle of the humerus, ulnar styloid, head of the third metacarpal, anterior superior iliac spine, posterior superior iliac spine, greater trochanter, lateral femoral condyle, medial femoral condyle, lateral malleolus, medial malleolus, calcaneus and base of the 5th metatarsal. Three additional markers were placed on the baseball bat: one along the shaft near the grip and two more on opposite sides of the cap at the end of the barrel. An additional marker was placed on the batting tee to determine the instant of ball contact. Tee height and distance from the tee to the force plates was adjusted by each batter such that the ball was in the preferred position to be hit with maximal force.

Kinematic data were recorded with 10 infrared cameras (BTS Bioengineering, Quincy, MA) at 500 Hz. Ground reaction forces under the lead and trail feet were recorded at 1000 Hz with two separate force plates (BTS Bioengineering, Quincy, MA). The global coordinate system was set up such that positive Y was vertically upward and the X direction was orthogonal to Y and positive in the direction of the pitcher. The Z direction was the cross-product of Y and X. The motion data from the body markers were then filtered with a fourth-order Butterworth low-pass filter with a cut-off frequency of 12 Hz in order to eliminate any high-frequency noise (Fleisig et al., 2013). As suggested by Tabuchi et al. (2007), the positions of the bat markers were unfiltered to avoid any unwanted shifts in the bat velocity data. Linear bat speed was defined as the magnitude of the velocity of the marker on the cap of the bat barrel and expressed in metres per second (m/s).

A customised model for each hitter was built using Visual 3D (C-Motion, Inc., Rockville, MD, USA) to compute the angular rotations and velocities of the pelvis, trunk, thighs, shanks, feet, upper arms and forearms. Segmental coordinate systems were set up in similar fashion to the global coordinate system. Transverse plane motion was defined by rotations about Y-axis, sagittal plane motion was defined by rotations about the X-axis and frontal plane motion was defined by rotations about the Z-axis. Internal moments at the lead and trail hips and knees were calculated via inverse dynamics also using Visual 3D (C-Motion, Inc., Rockville, MD, USA) using the same segment coordinate conventions.

Hip-shoulder separation was defined as the transverse plane rotation of the trunk relative to the pelvis and expressed in degrees (°). Peak transverse plane rotational velocities of the pelvis and trunk were expressed in degrees per second (°/s). Ground reaction forces in the vertical, anterior/posterior (A/P) (i.e., X direction) and horizontal (i.e., Z direction) directions under the lead and trail feet were expressed as percent body weight (%BW). Internal joint moments were expressed as percent of body weight times height (% BW*HT).

As forward weight-shift and the sequential rotation of body segments ideally occurs following lead foot contact and bat speed ideally peaks at ball contact (Lau & Glossbrenner, 1984; Welch et al., 1995), the phase of interest during each swing was between lead foot contact and ball contact. Lead foot contact was defined at the first instant after lead foot lift-off when the ground reaction force under the lead foot exceeded 10N. Ball contact was defined as the instant the distance between the tee marker and the closest bat barrel marker in the X direction reached zero following lead foot contact (Dowling & Fleisig, 2016). Peak values for all kinematic (hip-shoulder separation, pelvis rotation velocity, trunk rotation velocity and linear bat speed) and kinetic (lead- and trail- foot ground reaction forces, lead and trail hip and knee moments) variables were determined between the events of lead foot contact and ball contact. Additionally, the timings of these peak values were expressed as percentages of the time between lead foot contact and ball contact. Average bat speed and biomechanical metrics were calculated for the three highest-speed swings for each participant.

The associations between bat speed and the individual peak kinematic, joint moment and ground reaction force variables were assessed using Pearson correlation coefficients. *R* values between 0.20 and 0.29 indicated a weak relationship, while *r* values between 0.30 and 0.39 indicated a moderate relationship and *r* values between 0.40 and 0.69 indicated a strong relationship. *R* values above 0.70 were indicative of a very strong relationship (Akoglu, 2018; Dancey & Reidy, 2004). *P* values less than 0.05 indicated significant correlations between linear bat speed at ball contact and each individual kinematic or kinetic variable. Stepwise multiple regression was then used to determine if combinations of independent variables improved the prediction of linear bat speed beyond any single predictive variable. For stepwise multiple regression analyses, variables that had a bivariate relationship with the dependent variable with a *P* value of 0.100 or less were included as factors in the multiple regression in a stepwise manner. The r^2 and R^2 values represented the percentage of the variability in the dependent variable (linear bat speed at ball contact) explained by independent variables (r^2) or combinations of variables (R^2). To compare timings of the biomechanical variables thought to associate with bat speed, Bonferroni-corrected pairwise t-tests were used to determine significant differences in the timings (expressed as a % of time from lead foot contact to ball contact) of 6 events: peak pelvis velocity, peak trunk velocity, peak hip-shoulder separation, peak lead leg GRF, peak lead knee extension moment and peak lead hip extension moment. The threshold for statistically different event times was $P < 0.05$. All statistics were performed using SPSS v. 21.0 software (IBM Corp., Chicago, IL).

Results

Average bat speed across all subjects was 30.0 ± 2.0 m/s. Average peak rotational velocities for the pelvis and trunk were 620 ± 43°/s and 794 ± 51°/s, respectively. Hip-shoulder separation at foot contact averaged 25 ± 11° and peak hip-shoulder separation was 41 ± 9° (Figure 1).

Peak vertical GRF under the lead foot was 159 ± 29% BW while the peak A/P GRF was 57 ± 12% BW and directed posteriorly (i.e., towards the catcher). Peak horizontal GRF was 21 ± 12% BW. Peak resultant lead foot GRF reached 170 ± 30% BW. Peak lead knee extension moment was 9.1 ± 2.9% BW*HT and the peak hip extension moment was 8.0 ± 2.7% BW*HT (Figure 2).

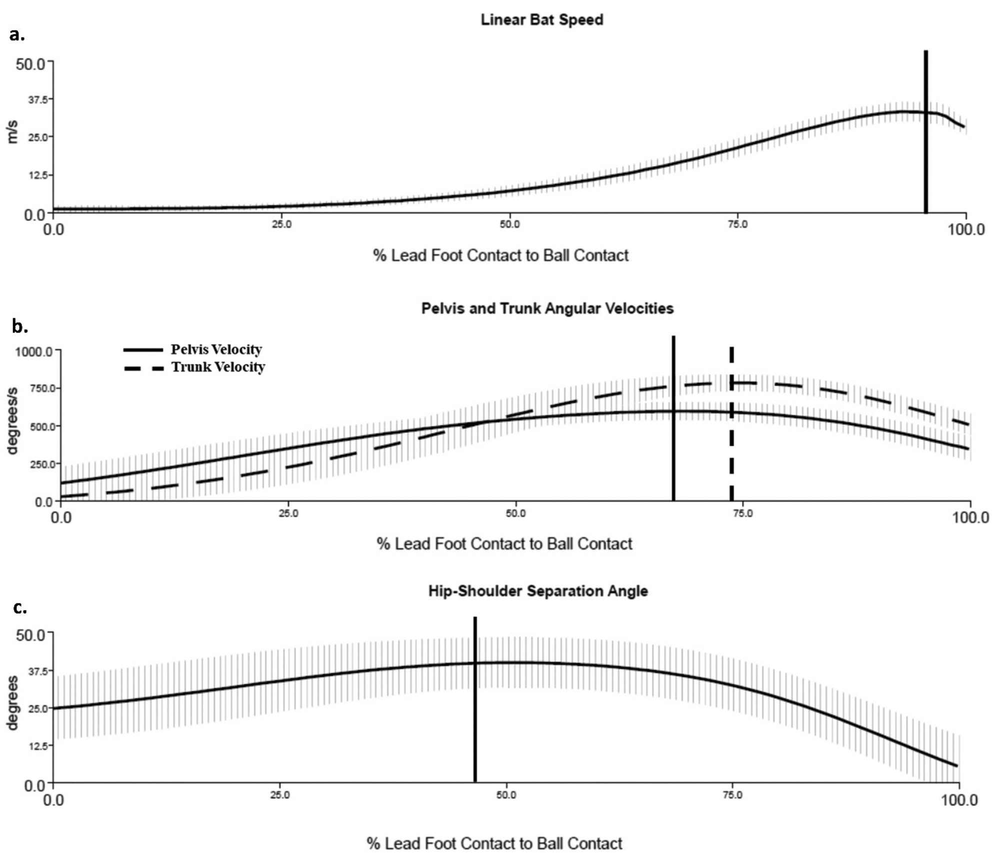

Figure 1. Ensemble averages of: (a) linear bat speed (m/s), (b) pelvis (solid line) and trunk (dashed line) transverse rotation velocities (degrees/s) and (c) hip-shoulder separation (degrees) between the events of lead foot contact (0%) to ball contact (100%). Vertical lines represent the times at which each variable reaches its respective peak.

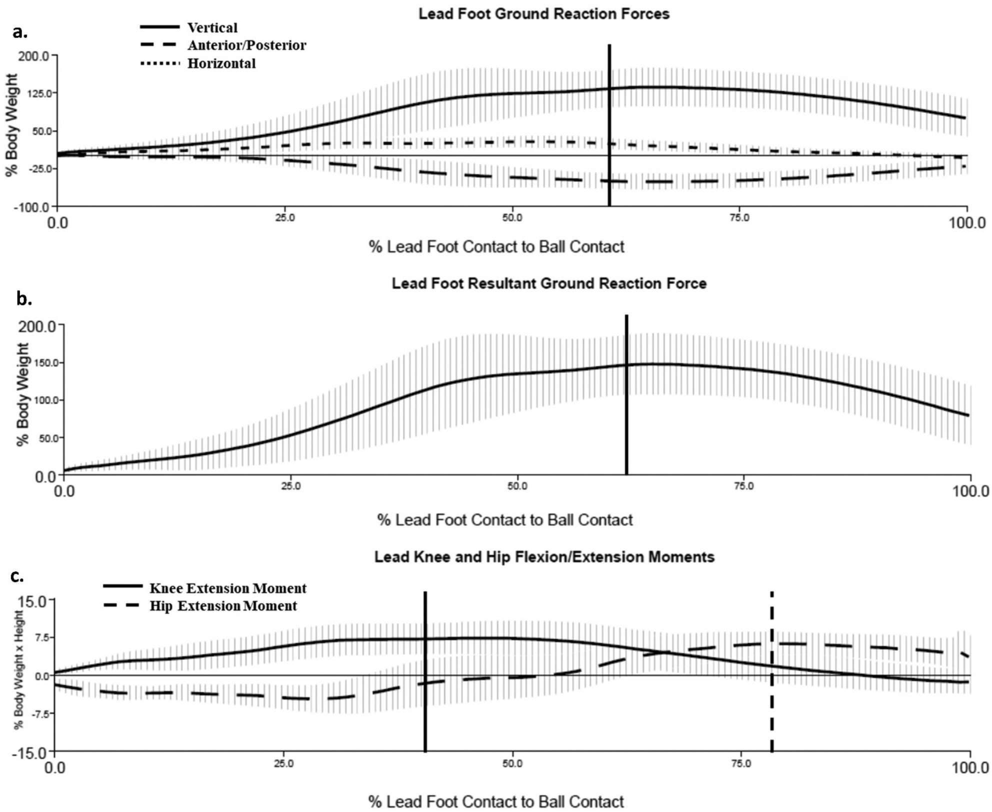

Figure 2. Ensemble averages of: (a) lead foot vertical (solid line), anterior/posterior (dashed line) and horizontal (dotted line) ground reaction forces, (b) lead leg resultant ground reaction force and (c) lead knee extension moment (solid line) and lead hip extension moment (dashed line) between the events of lead foot contact (0%) to ball contact (100%). Ground reaction forces are expressed as a percentage of body weight (% BW) and the joint moments are expressed as a percentage of body weight times height (% BW*HT). Vertical lines represent the times at which each variable reaches its respective peak. Positive values represent internal knee extension, hip extension and hip abduction moments.

Peak vertical lead foot GRF (r = 0.662, $P = 0.001$) and peak lead foot A/P GRF (r = 0.574, $P = 0.008$) were both strongly correlated to bat speed. Peak resultant GRF under the lead foot was also strongly correlated to bat speed (r = 0.662, $P = 0.001$). Peak lead knee extension moment and peak lead hip extension moment displayed moderate and nonsignificant associations with bat speed (r = 0.398 and r = 0.383, respectively, $P > 0.05$). However, both were entered into the multiple regression based on their independent relationships to linear bat speed having a P value < 0.100. Additionally, peak knee extension moment was strongly correlated with both the peak vertical lead foot GRF (r = 0.496, $P = 0.026$) and the peak A/P lead foot GRF (r = 0.548, $P = 0.012$). Peak pelvis velocity (r = −0.09, $P = 0.706$), peak trunk velocity (r = −0.074, $P = 0.757$) and peak hip-shoulder separation (r = 0.034, $P = 0.888$) all displayed negligible and nonsignificant relationships to bat speed. No combination of independent variables strengthened the relationship to bat speed beyond what these variables predicted on their own. (Tables 1–3)

Table 1. Correlations between bat speed at ball contact and pelvis and Trunk Kinematics.

Factor	Mean (SD)	Correlation Coeff.	% Explained	*P*-Value
Hip-Shoulder Separation at Lead Foot Contact (°)	25 (11)	0.064	0.4%	0.788
Peak Hip-Shoulder Separation (°)	41 (9)	0.034	0.1%	0.888
Peak Pelvis Velocity (°/s)	620 (43)	−0.090	0.8%	0.706
Peak Trunk Velocity (°/s)	794 (51)	−0.074	0.5%	0.757

Table 2. Correlations between bat speed at ball contact and lead and trail ground reaction forces.

Factor	Mean (SD)	Correlation Coeff.	% Explained	*P*-Value	Regression Equation
Peak Vertical GRF (%BW)					
Lead	159 (29)	0.622	39%	**0.001**	**0.050× + 21.8**
Trail	71 (16)	−0.270	7.3%	0.249	
Peak A/P GRF (%BW)					
Lead	−57 (12)	−0.574	33%	**0.008**	**−0.11× + 23.8**
Trail	21 (6)	0.194	3.8%	0.414	
Peak Horizontal GRF (%BW)					
Lead	21 (12)	0.138	1.9%	0.562	
Trail	25 (4)	−0.315	10%	0.176	
Peak Resultant GRF (%BW)					
Lead	170 (30)	0.622	39%	**0.001**	**0.047× + 21.7**
Trail	73 (19)	−0.056	0.3%	0.816	

*Regression equation only included for correlations included in the multiple regression ($P < 0.1$).

Table 3. Correlations between bat speed at ball contact and lead and trail knee and hip kinetics.

Factor	Mean (SD)	Correlation Coeff.	% Explained	*P*-Value	Regression Equation
Peak Knee Extension Moment (%BW*HT)					
Lead	9.1 (2.9)	0.398	16%	0.082	0.30× + 27.0
Trail	3.6 (2.8)	−0.192	3.7%	0.418	
Peak Hip Extension Moment (%BW*HT)					
Lead	8.0 (2.7)	0.383	15%	0.096	0.31× + 27.3
Trail	13.2 (2.1)	0.220	5%	0.350	
Peak Hip Abduction Moment (%BW*HT)					
Lead	5.5 (2.0)	0.184	3.4%	0.438	
Trail	5.4 (2.9)	−0.015	0.02%	0.951	
Peak Hip Adduction Moment (%BW*HT)					
Lead	7.0 (2.8)	−0.003	0.0009%	0.991	
Trail	4.4 (3.2)	0.244	6%	0.299	
Peak Hip External Rotation Moment (%BW*HT)					
Lead	2.0 (0.9)	0.284	8%	0.224	
Trail	2.8 (0.8)	0.175	3%	0.461	
Peak Hip Internal Rotation Moment (%BW*HT)					
Lead	1.7 (1.0)	−0.060	0.4%	0.803	
Trail	0.6 (1.1)	−0.146	2.1%	0.540	

*Regression equation only included for correlations included in the multiple regression ($P < 0.1$).

The results of the pairwise comparisons show the sequential order of kinetic and kinematic events from lead foot contact to ball contact (Figure 3, Table 4). Following lead foot contact, peak lead knee extension moment and peak hip-shoulder separation occurred nearly simultaneously at 44 ± 10% and 47 ± 12% of the time from foot contact to ball contact and significantly earlier than all other events ($P < 0.001$ compared to all other events). Peak lead foot GRF and peak pelvis velocity also occurred nearly simultaneously at 62 ± 14% and 68 ± 8%, respectively. Additionally, both of these variables (peak

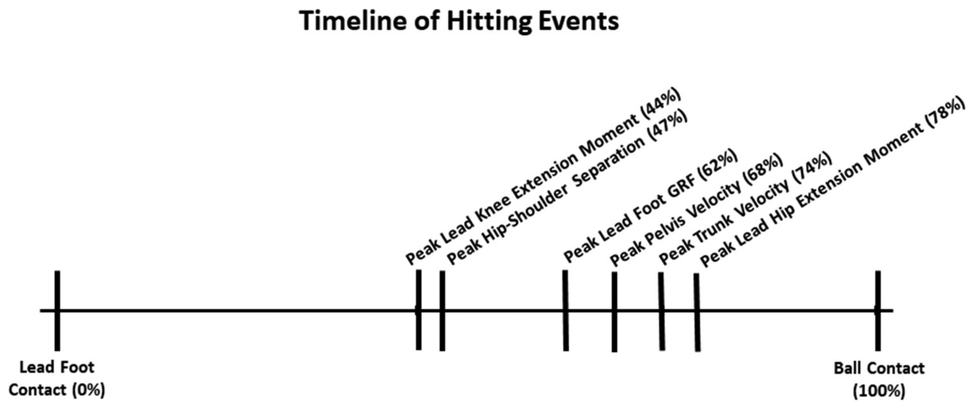

Figure 3. Timing of peak kinematic and kinetic variables as a percentage of the time from lead foot contact (0%) to ball contact (100%).

Table 4. Pairwise comparisons of timings of hitting events.

	Pk. Lead Knee Ext. Moment	Pk. Hip-Shoulder Separation	Pk. Lead Leg GRF	Pk. Pelvis Velocity	Pk. Trunk Velocity	Pk. Lead Hip Ext. Moment
Pk. Lead Knee Ext. Moment						
Mean Difference		−3%	−18%	−24%	−30%	−34%
(95% C.I.)		(−10, 4)	(−26, −11)	(−29, −19)	(−35, −26)	(−39, −30)
P-Value		0.378	**<0.001**	**<0.001**	**<0.001**	**<0.001**
# of Hitters		12/20	16/20	20/20	20/20	20/20
Pk. Hip-Shoulder Separation						
Mean Difference	−3%		−15%	−21%	−27%	−31%
(95% C.I.)	(−10, 4)		(−20, −11)	(−27, −16)	(−32, −23)	(−39, −23)
P-Value	0.378		**<0.001**	**<0.001**	**<0.001**	**<0.001**
# of Hitters	12/20		19/20	19/20	20/20	20/20
Pk. Lead Leg GRF						
Mean Difference	−18%	−15%		−6%	−12%	−16%
(95% C.I.)	(−26, −11)	(−20, −11)		(−11, −0.2)	(−17, −7)	(−24, −8)
P-Value	**<0.001**	**<0.001**		0.210	**<0.001**	**<0.001**
# of Hitters	16/20	19/20		17/20	17/20	17/20
Pk. Pelvis Velocity						
Mean Difference	−24%	−21%	−6%		−6%	−10%
(95% C.I.)	(−29, −19)	(−27, −16)	(−11, −0.2)		(−9, −3)	(−16, −5)
P-Value	**<0.001**	**<0.001**	0.210		**<0.001**	**0.001**
# of Hitters	20/20	19/20	17/20		17/20	17/20
Pk. Trunk Velocity						
Mean Difference	−30%	−27%	−12%	−6%		−4
(95% C.I.)	(−35, −26)	(−32, −23)	(−17, −7)	(−9, −3)		(−10, 2)
P-Value	**<0.001**	**<0.001**	**<0.001**	**<0.001**		0.151
# of Hitters	20/20	20/20	17/20	17/20		12/20
Pk. Lead Hip Ext. Moment						
Mean Difference	−34%	−31%	−16%	−10%	−4	
(95% C.I.)	(−39, −30)	(−39, −23)	(−24, −8)	(−16, −5)	(−10, 2)	
P-Value	**<0.001**	**<0.001**	**<0.001**	**0.001**	0.151	
	20/20	20/20	17/20	17/20	12/20	
# of Hitters						

lead foot GRF and peak pelvis velocity) were significantly different from all other events ($P < 0.001$). Lastly, there was no significant difference in the timing of peak trunk velocity and peak lead hip extension moment (74 ± 4% and 78 ± 11%, respectively, $P = 0.151$). However, both occurred significantly later compared to previous events ($P < 0.001$, compared to all previous events).

Discussion and implications

The results of this study demonstrate strong associations between peak ground reaction forces under the lead foot and bat speed in collegiate hitters. Bat speed increases as the peak lead foot vertical or posterior GRF increases. Based on the mean data and the regression equations, increasing the lead foot peak vertical or posterior GRF by 10% would predict an increase in bat speed by 1.8% and 2.3%, respectively.

Lead foot ground reaction forces have also been associated with ball velocity during pitching. Previous research has demonstrated the importance of the lead leg in generating the necessary posterior, or 'braking', forces to convert the forward momentum of the body into rotational energy, which is then transferred through the kinetic chain to generate ball velocity (Guido & Werner, 2012; Guido et al., 2009; MacWilliams et al., 1998). The results of the current study indicate that a similar mechanism could be used to generate bat speed. Following lead foot contact, the lead foot posterior ground reaction force acts to slow the forward momentum of the body while driving the rotation of the pelvis towards the pitcher. The rotation of the pelvis, while the trunk and shoulders are still rotated away from the pitcher (i.e., in the coiled position) increases the hip-shoulder separation, stretches the abdominal musculature and activates the stretch-shortening cycle. The elastic recoil of these muscles, along with the peaking lead leg ground reaction forces, accelerates the pelvis and trunk with peak pelvis velocity occurring first, followed by peak trunk velocity and finally, peak bat speed. Additionally, peak resultant GRF under the lead foot exceeded body weight by 70%, on average, indicating that there was additional force being applied to the ground beyond the transfer of body weight from the trail foot to the lead foot. This may have been achieved by extension of the lead knee shortly after foot contact, as peak lead knee extension moment occurred significantly before peak lead foot GRF and was significantly correlated to both the vertical and A/P components.

Many of the biomechanical and performance variables of hitting in the current study compared favourably with those of previous studies. Fortenbaugh et al. (Fortenbaugh et al., 2011). found similar the peak magnitudes of the lead foot posterior (i.e., 'braking') and vertical ground reaction forces (−57 ± 5% BW and 150 ± 13% BW, respectively) compared to the current study (−57 ± 12% BW and 159 ± 29% BW, respectively). However, it should be noted that the subjects in this study were professional hitters swinging at fastballs from live pitchers. Ae et al. (Ae et al., 2018) also examined collegiate baseball players hitting off of a tee and reported a higher bat speed (37.4 m/sec) compared to the current study. However, it is not known if this occurred at ball contact or before. Additionally, hitters in this study also generated higher peak lead knee extension moments compared to those in the current study (about 11.6% BW*HT vs. 9.1% BW*HT) and higher peak lead hip extension moments (about 11.6% BW*HT, compared to 8.0% BW*HT). Finally, Escamilla et al. (Escamilla et al., 2009) found similar linear bat speed at ball contact (30.0 ± 2 m/sec) to the current study, although peak bat speed was not reported.

Traditionally, training to increase bat speed has included drills to increase hand speed and rotational velocity (Szymanski et al., 2006, 2007). For coaches and players seeking to improve bat velocity, improving lower extremity strength and the ability to transfer ground reaction forces through the kinetic chain may also be beneficial. As evidenced by the associations between squat strength and linear bat speed (Szymanski et al., 2010), the use of closed kinetic chain exercises to strengthen the muscles around the knee and hip seems likely to improve lower extremity kinetics and joint moments following lead foot contact, which could lead to increases in bat speed. Additionally, hitting-specific drills which force the hitter to stabilise the lead leg and push into the ground quickly after foot contact may also serve a similar purpose. Two common drills used to train proper lead leg mechanics are the flamingo and the K-posture drills. During the flamingo drill, the hitter begins with an elevated lead foot and all of the body weight on the trail foot. From this position, the batter strides forward, shifting weight to the lead foot, and swings the bat. The K-posture drill positions the trail foot on a 45° slant board, which increases the load on the trail foot during the batter’s stance. Following lead foot contact, the increased load is transferred to the lead leg and through the kinetic chain. Future studies examining the effects of these drills on ground reaction forces, lead leg mechanics and bat speed are needed.

While the current study highlights the association of lead leg ground reaction forces and hitting performance, there are a few limitations that warrant mention. First, ball exit velocity, which is a more commonly used indicator of batting performance, could not be measured. Due to space constraints in the laboratory, the distance from the tee to the net was not large enough to measure exit velocity with a radar gun. Thus, linear bat speed at ball contact was used as the primary dependent variable. Second, all of the participants in this study used the bat that was provided instead of a bat of their preferred length and weight. This may explain why peak bat speed did not occur a ball contact for most hitters as peak bat speed may have been achieved earlier using a lighter than normal bat. An additional reason for peak bat speed occurring before ball impact in this group may be the fact that hitting mechanics were measured before the season and regular practice of hitting skills had begun. Regular batting practice before biomechanical assessment may allow for the hitters to regain their optimal swing mechanics such that peak bat speed is reached at ball contact. Third, the hitters hit the ball off of a tee instead of a pitch thrown by a live pitcher or a pitching machine. Previous studies have shown significant differences in segment kinematics and joint kinetics in batters hitting a thrown ball versus hitting a stationary ball off of a tee (Ae et al., 2018; Washington & Oliver, 2018). This could also explain the short delay between peak bat speed and ball contact. Additionally, swing mechanics and timing can also be altered due to pitch speed and location (Fortenbaugh et al., 2011). Therefore, while batting from a tee was not reflective of an in-game situation for each batter, it did eliminate variables such as pitch speed and location in order to ensure consistency from swing to swing. Finally, as only collegiate baseball players were examined in the current study, it is not known if the results are generalisable to batters at other levels of competition (i.e., youth or professionals).

Conclusion

This study highlights the association between lead leg kinetics and bat speed as greater force production under the lead foot in the vertical or posterior directions predicted increased bat speed. Training to improve bat speed should include closed kinetic chain exercises aimed at improving the overall strength and power in the muscle groups of the hip and knee. Batting-specific exercises should also be implemented to improve lead leg mechanics and emphasise the application of force into the ground. Future research should explore the effects of these types of lower extremity training on bat speed.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

References

Ae, K., Koike, S., Fujii, N., Ae, M., Kawamura, T., & Kanahori, T. (2018). A comparison of kinetics in the lower limbs between baseball tee and pitched ball batting. *Human Movement Science*, *61*, 126–134. https://doi.org/10.1016/j.humov.2018.07.010

Akoglu, H. (2018). User's guide to correlation coefficients. *Turkish Journal of Emergency Medicine*, *18*(3), 1–93. https://doi.org/10.1016/j.tjem.2018.08.001

Dancey, C. P., & Reidy, J. (2004). *Statistics without maths for psychology*. Pearson Education.

DeRenne, C. (2011). *The scientific approach to hitting*. University Readers.

Dowling, B., & Fleisig, G. S. (2016). Kinematic comparison of baseball batting off of a tee among various competition levels. *Sports Biomechanics*, *15*(3), 255–269. https://doi.org/10.1080/14763141.2016.1159320

Escamilla, R. F., Fleisig, G. S., DeRenne, C., Taylor, M. K., Moorman, C. T., Imamura, R., Barakatt, E., & Andrews, J. A. (2009). A comparison of age level on baseball hitting kinematics. *Journal of Applied Biomechanics*, *25*(3), 210–218. https://doi.org/10.1123/jab.25.3.210

Fleisig, G. S., Hsu, W. K., Fortenbaugh, D., Cordover, A., & Press, J. M. (2013). Trunk axial rotation in baseball pitching and batting. *Sports Biomechanics*, *12*(4), 324–333. https://doi.org/10.1080/14763141.2013.838693

Fortenbaugh, D., Fleisig, G., Onar-Thomas, A., & Asfour, S. (2011). The effect of pitch type on ground reaction forces in the baseball swing. *Sports Biomechanics*, *10*(4), 270–279. https://doi.org/10.1080/14763141.2011.629205

Guido, J. A., Jr., & Werner, S. L. (2012). Lower-extremity ground reaction forces in collegiate baseball pitchers. *Journal of Strength & Conditioning Research*, *26*(7), 1782–1785. https://doi.org/10.1519/JSC.0b013e31824e1211

Guido, J. A., Jr., Werner, S. L., & Meister, K. (2009). Lower-extremity ground reaction forces in youth windmill softball pitchers. *Journal of Strength & Conditioning Research*, *23*(6), 1873–1876. https://doi.org/10.1519/JSC.0b013e3181b42535

Han, K. H., Como, C., Kim, J., Lee, S., Kim, J., Kim, D. K., & Kwon, Y. H. (2019). Effects of the golfer-ground interaction on clubhead speed in skilled male golfers. *Sports Biomechanics*, *18*(2), 115–134. https://doi.org/10.1080/14763141.2019.1586983

Inkster, B., Murphy, A., Bower, R., & Watsford, M. (2011). Differences in the kinematics of the baseball swing between hitters of varying skill. *Medicine & Science in Sports and Exercise*, *43*(6), 1050–1054. https://doi.org/10.1249/MSS.0b013e318203626a

Joyce, C. (2017). The most important "factor" in producing clubhead speed in golf. *Human Movement Science*, *55*(10), 138–144. https://doi.org/10.1016/j.humov.2017.08.007

Lau, C., & Glossbrenner, A. (1984). *The winning hitter: How to play championship baseball*. Hearst.

MacWilliams, B. A., Choi, T., Perezous, M. K., Chao, E. Y., & McFarland, E. G. (1998). Characteristic ground-reaction forces in baseball pitching. *American Journal of Sports Medicine*, *26*(1), 66–71. https://doi.org/10.1177/03635465980260012801

McNally, M. P., Yontz, N., & Chaudhari, A. M. (2014). Lower extremity work is associated with club head velocity during the golf swing in experienced golfers. *International Journal of Sports Medicine*, *35*(9), 785–788. https://doi.org/10.1055/s-0034-1367010

Putnam, C. (1993). Sequential motions of body segments in striking and throwing skills: Descriptions and explanations. *Journal of Biomechanics*, *26*, 125–135. https://doi.org/10.1016/0021-9290(93)90084-R

Shaffer, B., Jobe, F. W., Pink, M., & Perry, J. (1993). Baseball batting: An electromyographic study. *Clinical Orthopaedics & Related Research*, *292*, 285–293. https://doi.org/10.1097/00003086-199307000-00038

Szymanski, D. J., McIntyre, J. S., Szymanski, J. M., Bradford, T. J., Schade, R. L., Madsen, N. H., & Pascoe, D. D. (2007). Effect of 12 weeks of torso rotational strength on angular hip, angular shoulder, and linear bat swing velocities of high school baseball players. *Journal of Strength & Conditioning Research*, *21*(4), 1117–1125. https://doi.org/10.1519/00124278-200711000-00024

Szymanski, D. J., McIntyre, J. S., Szymanski, J. M., Molloy, J. M., Madsen, N. H., & Pascoe, D. (2006). Effect of wrist and forearm training on linear bat-end, center of percussion, and hand velocities, and time to contact of high school baseball players. *Journal of Strength & Conditioning Research*, *20*(1), 231–240. https://doi.org/10.1519/00124278-200602000-00038

Szymanski, D. J., Szymanski, J. M., Schade, R. L., Bradford, T. J., McIntyre, J. S., DeRenne, C., & Madsen, N. H. (2010). The relation between anthropometric and physiological variables and bat velocity of high-school baseball players before and after 12 weeks of training. *Journal of Strength & Conditioning Research*, *24*(11), 2933–2943. https://doi.org/10.1519/JSC.0b013e3181f0a76a

Tabuchi, N., Matsuo, T., & Hashizume, K. (2007). Bat speed, trajectory, and timing for collegiate baseball batters hitting a stationary ball. *Sports Biomechanics*, *6*(1), 17–30. https://doi.org/10.1080/14763140601058409

Washington, J., & Oliver, G. (2018). Kinematic differences between hitting off a tee versus front toss in collegiate softball players. *International Biomechanics*, *5*(1), 30–35. https://doi.org/10.1080/23335432.2018.1472038

Welch, C. M., Banks, S. A., Cook, F. F., & Draovitch, P. (1995). Hitting a baseball: A biomechanical description. *JOSPT*, *22*(5), 193–201. https://doi.org/10.2519/jospt.1995.22.5.193

Williams, T., & Underwood, J. (1970). *The science of hitting*. Simon & Schuster.